Earn Your Wings

Applied Machine Learning for Systematic Crypto

Build a machine learning signal in CrowdCent Cloud and submit it to the CrowdCent Challenge for a t-shirt and pin. Keep improving and experimenting with your existing models



DECENTRALIZED AI DAY NEW YORK

Hosted by the Numerai Council of Elders

Jason Rosenfeld Co-founder & CEO, CrowdCent

10 OCT 2026

Order of operations

15:45	**Submit your first model**	Open Cloud, run the notebook, submit	15 MIN
16:00	**How you're scored**	Spearman, NDCG@40, then uniqueness against the meta-model	5 MIN
16:05	**Make it yours with Centaur**	Your own idea, ask Centaur, validate, overwrite slot 1	20 MIN
16:25	**Keep it submitting on a schedule**	Schedule to run on every inference data release	5 MIN
16:30	**If there's time: pick a recipe**	Optuna tuning, simulate the meta-model, track your performance	5 MIN
16:35	**Homework: the Ultimate Submission**	Three data sources, four models, one ensemble, then where to go next	5 MIN

You'll need a laptop with a browser. Optionally, for Numerai Crypto you'll need a numer.ai account and an API key.

What is CrowdCent Cloud?



- An IDE in your browser, and in the cloud.
- A persistent notebook and an object store.
- A scheduled pipeline and a home for your models.
- An AI agent, Centaur, in the sidebar.

While it plays: open crowdcent.com/cloud and sign up

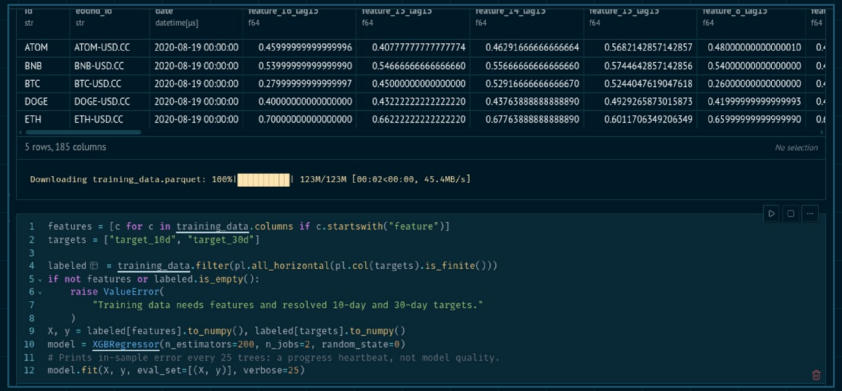
Open CrowdCent Cloud and fork the starter project

1 crowdcent.com/cloud → Sign up and submit free.

2 The notebook opens: Hyperliquid ranking, end to end. It starts running in your browser on its own.

3 Check Challenge access is on, top right. The notebook gets a short-lived key, so there is no key to paste.

4 It downloads about 125 MB and trains in a few minutes. Leave the tab open.



id	ecosia_id	date	feature_10_lag1	feature_13_lag1	feature_14_lag1	feature_17_lag1	feature_6_lag1	feat
str	str	datetime[μs]	f64	f64	f64	f64	f64	f64
ATOM	ATOM-USD.CC	2020-08-19 00:00:00	0.45999999999999996	0.40777777777777774	0.46291666666666664	0.5682142857142856	0.4800000000000010	0.4
BNB	BNB-USD.CC	2020-08-19 00:00:00	0.5399999999999990	0.5466666666666660	0.5566666666666660	0.5744642857142856	0.5400000000000000	0.4
BTC	BTC-USD.CC	2020-08-19 00:00:00	0.27999999999999997	0.45000000000000000	0.52916666666666670	0.5244047619047618	0.2600000000000000	0.4
DOGE	DOGE-USD.CC	2020-08-19 00:00:00	0.40000000000000000	0.43222222222222220	0.43763888888888890	0.4929265873015873	0.4199999999999993	0.4
ETH	ETH-USD.CC	2020-08-19 00:00:00	0.70000000000000000	0.66222222222222220	0.67763888888888890	0.6011706349206349	0.6599999999999990	0.4

5 rows, 185 columns — No selection

Downloading training_data.parquet: 100%|██████████| 123M/123M [00:02<00:00, 45.4MB/s]

```
features = [c for c in training_data.columns if c.startswith("feature")]
targets = ["target_10d", "target_30d"]

labeled = training_data.filter(pl.all_horizontal(pl.col(targets).is_finite()))
if not features or labeled.is_empty():
    raise ValueError(
        "Training data needs features and resolved 10-day and 30-day targets."
    )
X, y = labeled[features].to_numpy(), labeled[targets].to_numpy()
model = XGBRegressor(n_estimators=200, n_jobs=2, random_state=0)
# Prints in-sample error every 25 trees: a progress heartbeat, not model quality.
model.fit(X, y, eval_set=[(X, y)], verbose=25)
```

Notebook won't run? Slide 07.

What the starter project is doing

hyperliquid_ranking.py

```python
client = cc.ChallengeClient("hyperliquid-ranking")
client.download_training_dataset("latest",
"training_data.parquet")
model = XGBRegressor(n_estimators=200).fit(X, y)

client.download_inference_data("latest", "inference_data.csv")
predictions = model.predict(live_X)
client.submit_predictions(df=predictions, slot=1)

# X: the feature_* columns   y: target_10d, target_30d
```

training_data.parquet

id	date	feature_1_lag0	…	feature_45_lag0	target_10d	target_30d
BTC	2026-09-08	0.454	…	0.249	0.260	0.320
ETH	2026-09-08	0.464	…	0.320	0.420	0.201
SOL	2026-09-08	0.420	…	0.243	0.550	0.391
HYPE	2026-09-08	0.469	…	0.414	0.586	0.195
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Submit id, pred_10d and pred_30d, each a float in [0, 1], for at least 80 ids. You get five slots per day.

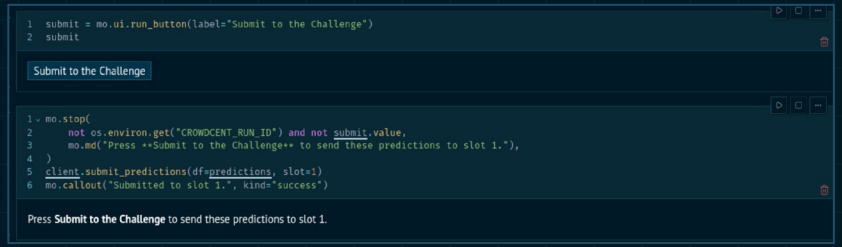
- 169 Hyperliquid perps, one row per perp per day, history back to August 2020.
- 180 features: 45 ranked signals at lags 0, 5, 10 and 15. Two targets: 10- and 30-day forward returns, ranked.

predictions.csv

id	pred_10d	pred_30d
UNI	0.696	0.764
BNB	0.630	0.762
HYPE	0.571	0.748
⋮	⋮	⋮

Press Submit, then collect your wings.

1. Press Submit to the Challenge. You'll see “Submitted to slot 1.”
2. The live submission window is closed this afternoon, but it queues and goes in when the next period opens at 14:00 UTC (10:00am ET).
3. Show a helper your queued submission.





T-SHIRT AND PIN

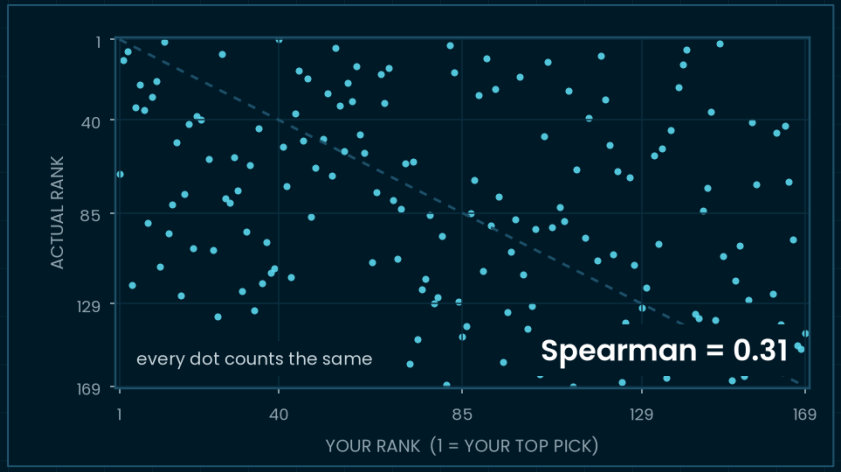
Stuck? Submit from anywhere you have Python.

crowdcent.py

```
import crowdcent_challenge as cc, polars as pl   # key: CROWDCENT_API_KEY from crowdcent.com/profile/settings
client = cc.ChallengeClient("hyperliquid-ranking")
client.download_inference_data("latest", "live.parquet")
live = pl.read_parquet("live.parquet")
score = pl.mean_horizontal(pl.col("^feature_.*_lag0$")).rank() / pl.len()
client.submit_predictions(df=live.select("id", pred_10d=score, pred_30d=score))
```

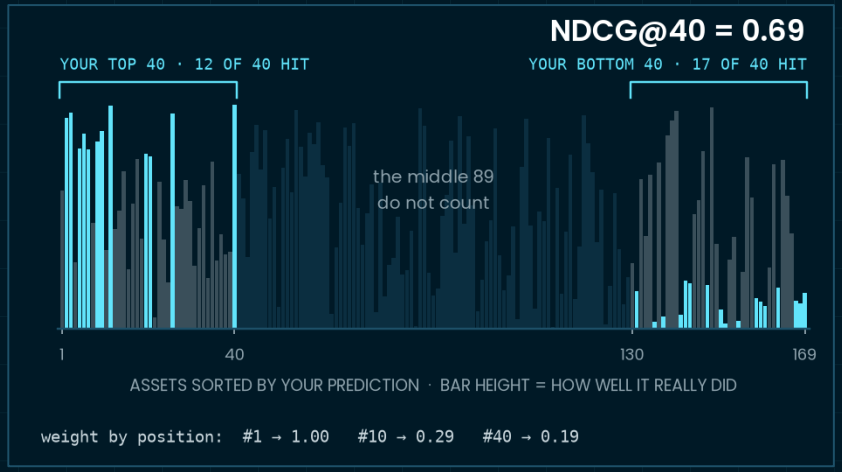
- pip install crowdcent-challenge polars, export your API key, run it. A few seconds, no 125 MB download, no training.
- It ranks assets on the average of their lag-0 features: a placeholder that counts for the t-shirt and pin. Swap the score line for your model tonight.

Spearman scores the rank order
NDCG@40 cares more about the tails.



SPEARMAN

Rank correlation across the whole universe. Every position counts equally.



NDCG@40

How well you order the top 40 and the bottom 40. Random scores about 0.55, not 0.5.

SYNTHETIC DATA, FOR ILLUSTRATION

Scores come in daily from Hyperliquid prices.

UNIQUENESS

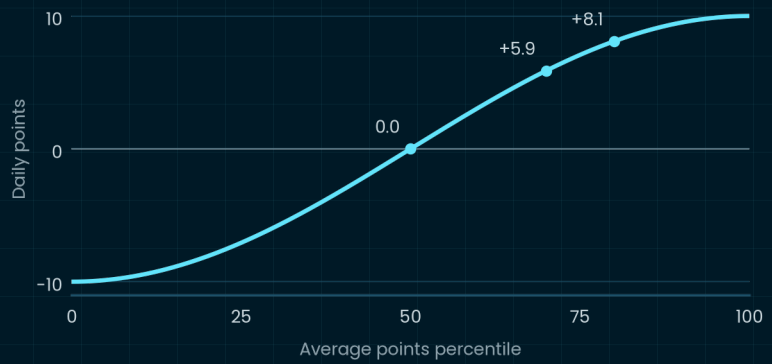
Your predictions are neutralized against the meta-model, then scored again. New signal is what counts.

YOUR PREDICTIONS − FIT TO META-MODEL (β · meta + α, least squares) = RESIDUAL

SCORED AGAIN: UNIQUE SPEARMAN, UNIQUE NDCG@40

scoring

```
from crowdcent_challenge.scoring import evaluate_hyperliquid_submission
```



- Your percentile score is how well you do compared to everybody in each round
- You earn CC Points based on your percentile

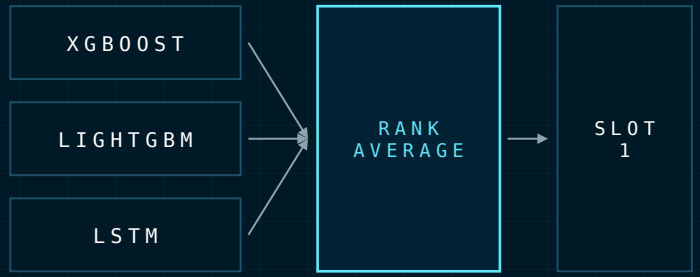
  $$\text{CC Points} = -\cos(\text{percentile} \times \pi / 100) \times 10$$

- CC Points set your weight in the meta-model and your tier

Everyone should build different models.

features.py

```
train = training_data.with_columns(
    ((pl.col(f"feature_{i}_lag0") - pl.col(f"feature_{i}_lag15"))
     .rank().over("date") /
pl.len().over("date")).alias(f"feature_{i}_delta")
    for i in range(1, 46)
)
```



01 Features

The four lags are a short time series per signal. Re-rank the change per date, and apply the same step to the inference file.

02 Models

Train two or three different learners and average their ranks per date, not their raw outputs. centimators' LSTMRegressor reads the 4 lags as a sequence.

03 Data, tonight

Join Numerai Crypto's features or YIEDL's on symbol and date. The multi-source tutorial in the docs does the merge.

your own idea Your own idea first. Stuck? Paste the prompt for your birth month into Centaur:

```
Jan–Mar  "Add a trend feature per signal: lag0 minus lag15, ranked per date. Retrain with them."
Apr–Jun  "Add a linear model (Ridge) and blend it with XGBoost by averaging their ranks per date."
Jul–Sep  "Try max_depth 3, 6 and 9, scored on the last 90 days after a 30-day gap. Keep the best."
Oct–Dec  "Which features matter most? Retrain on the top 60 only. Does it beat the baseline?"
```

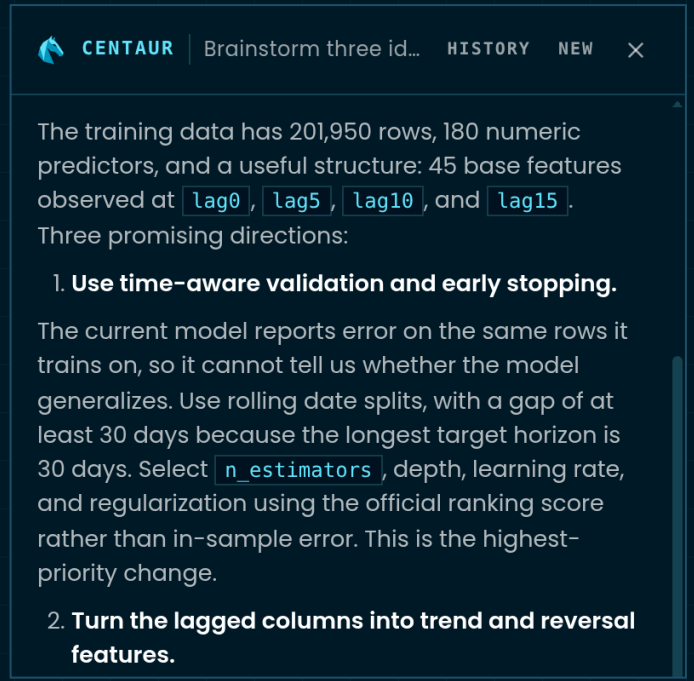
Ask Centaur for ideas, in the Cloud sidebar.

1. If this was your first submission, reload the notebook after you submit. Centaur (Ask AI) will now appear at the top right.
2. Accept the one-time Work with Centaur notice. It needs a verified email.
3. Paste some variation of this and send:

prompt

```
Brainstorm three ideas for improving this model from the
columns in the training data or the model type or anything
else!
```

Add your own idea at the end, or your birth-month prompt from slide 10 if you're stuck.



Check if it makes sense, then overwrite slot 1.

1 Ask Centaur to check both models on dates they never trained on:

prompt

```
Score my baseline and the new model walk-forward: folds by
date, a 30-day gap, scored per date. Which one wins?
```

Split by date, with a gap as long as the 30-day target. Shuffled folds leak tomorrow's returns into today's model.

2 If the new one wins, run the notebook to the end and press Submit to the Challenge again.

3 It goes to slot 1 and replaces your queued submission. One queued submission per slot; the newest one counts.

If it loses, keep the baseline and try another idea.

```
77
78  cv_results = pl.DataFrame(_cv_rows)
79  cv_summary = (
80      cv_results
81      .group_by("horizon")
82      .agg(
83          pl.len().alias("folds"),
84          pl.col("mean_daily_spearman").mean().alias("mean_cv_spearman"),
85          pl.col("mean_daily_spearman").std().alias("std_cv_spearman"),
86          pl.col("best_iteration").mean().alias("mean_best_iteration"),
87      )
88      .sort("horizon")
89  )
90  cv_summary
```

Search...

horizon str	folds u32	mean_cv_spearman f64
target_10d	3	0.14689162055970065
target_30d	3	0.18854190752506547

2 rows, 5 columns

Optional: a few more lines to submit to Numerai Crypto.

	HYPERLIQUID RANKING	NUMERAI CRYPTO
UNIVERSE	169 Hyperliquid perps	300 tokens by market cap
KEY	id (kPEPE)	symbol (PEPE)
TARGETS	10d and 30d ranks	30-day return, 5 bins
COLUMNS	id, pred_10d, pred_30d	symbol, signal
MINIMUM	80 ids	100 symbols

Append these after the CrowdCent submit cell. About 120 of today's 169 names are in Numerai's live universe.

two more cells

```
from numerapi import CryptoAPI
napi = CryptoAPI()  # reads NUMERAI_PUBLIC_ID / _SECRET_KEY
napi.download_dataset("crypto/v2.0/live.parquet",
                      "live.parquet")
live = pl.read_parquet("live.parquet")

sub = (predictions
  .with_columns(symbol=pl.col("id")
                .str.replace(r"^k([A-Z])", "$1"))
  .join(live.select("symbol"), on="symbol")
  .select("symbol",
          signal=pl.col("pred_30d").rank() / (pl.len() + 1)))
sub.write_csv("numerai.csv")

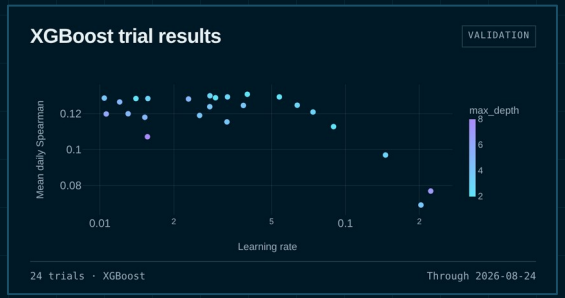
napi.upload_predictions("numerai.csv",
    model_id=napi.get_models()["your_model"])
```

Put it on a schedule so it keeps submitting.



RUN & SCHEDULE → ON RELEASE → SET SCHEDULE

If there’s time: pick another recipe from the cookbook.



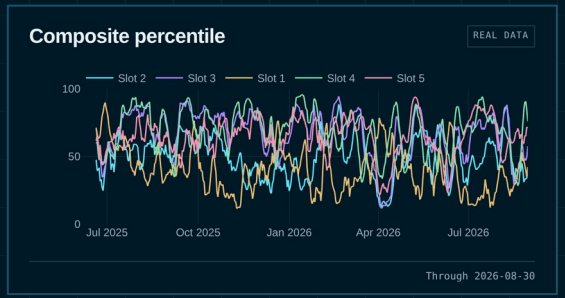
Tune a model with Optuna

Searches XGBoost settings, scored on mean daily Spearman over dates the model never saw. Then ask Centaur to put the best settings in your notebook and resubmit to slot 1.



Simulate the meta-model

Backtests the meta-model as a long/short book through the API, with the engine behind the Simulator. Knobs above your points tier are clamped.

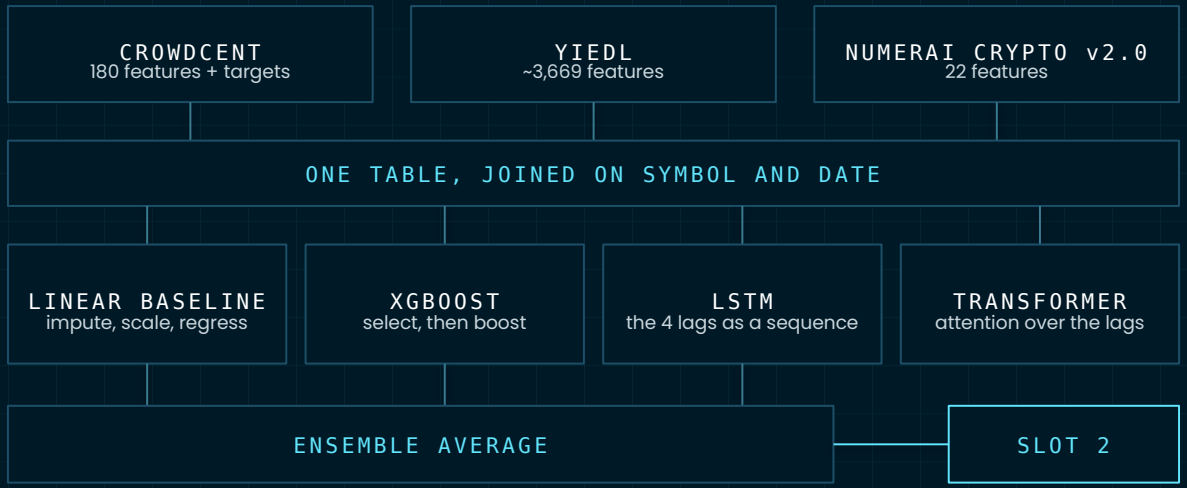


Track your performance

Every scored submission you've made, by slot, over time. Yours fills in once Sunday's round is scored.

CROWDCENT.COM/CLOUD/RECIPES

Tonight's challenge: the Ultimate Submission.



THE TUTORIAL

docs.crowdcent.com/tutorials/advanced-multi-data-slots/

WHAT IT NEEDS

A GPU. XGBoost runs on CUDA; the LSTM and Transformer are Keras models.

About 20 GB of disk while YIEDL downloads, 500 MB once cached. include_yiedl=False skips it.

About a 20-minute Cloud run: three folds, four models, then a retrain on everything.

Keep building after today.

CHALLENGE	crowdcent.com/challenge
DOCS	docs.crowdcent.com
CLOUD	crowdcent.com/cloud
DISCORD	discord.gg/v6ZSGuTbQS
COOKBOOK	crowdcent.com/cloud/recipes

CHALLENGE DOCS

NEXT IN THIS ROOM 17:00 Numeroo workshop with Ling